

//SIT 55 : ASSIGNMENT 10 : FACTORY DESIGN PATTERN

/*Implement Factory design pattern for the given context. Consider Car building process, which requires many steps from allocating accessories to final makeup. These steps should be written as methods and should be called while creating an instance of a specific car type. Hatchback, Sedan, SUV could be the subclasses of Car class. Car class and its subclasses, CarFactory and Test Factory Pattern should be implemented.*/

```
import java.util.Scanner;
abstract class Car_Factory
{
    String compnay,car_name;
    double budget;
    abstract void getprice(double price);
    abstract void detail(String company_name,String car_name);
    abstract void accessories();
    void input()
    {
        Scanner scan =new Scanner (System.in);
        System.out.print("Company- ");
        compnay=scan.next();
        System.out.print("Car- ");
        car_name=scan.next();
        System.out.print("Rough Budget(in Lakhs)- ");
        budget=scan.nextDouble();
    }
    void display(Car_Factory obj1)
    {
        obj1.getprice(budget);
        System.out.println("\n-----");
        obj1.detail(compnay, car_name);
        System.out.println("\n-----");
        obj1.accessories();
        System.out.println("\n-----");
    }
}
```

```
class Small_car extends Car_Factory
{
    String Ans;
    public void getprice(double price)
    {
        if(price>2&&price<5)
            Ans="No";
        else
            Ans="Yes";
    }
    public void detail(String company_name,String car_name)
    {
        System.out.println("Company- "+company_name);
        System.out.println("Name of Car- "+car_name);
        System.out.println("Color- Black/White/Orange/Red");
        System.out.println("Fuel- Petrol");
        System.out.println("Gears- Manual");
    }
    public void accessories()
```

```

{
    System.out.println("Types of Tyres- Alloy Wheels");
    System.out.println("Airbags- "+Ans);
    System.out.println("Back Wiper- "+Ans);
    System.out.println("Side Mirror- Two");
    System.out.println("Touch Screen Music Player- "+Ans);
}
}

```

```

class Sedan extends Car_Factory
{
    String Ans;
    public void getprice(double price)
    {
        if(price>6&&price<10)
            Ans="No";
        else
            Ans="Yes";
    }
    public void detail(String company_name,String car_name)
    {
        System.out.println("Company- "+company_name);
        System.out.println("Name of Car- "+car_name);
        System.out.println("Color- Black/White/Orange/Red");
        System.out.println("Fuel- Petrol/Diesel");
        System.out.println("Gears- Auto/Manual");
    }
    public void accessories()
    {
        System.out.println("Types of Tyres- Alloy Wheels");
        System.out.println("Airbags- YES");
        System.out.println("Back Wiper- YES");
        System.out.println("Side Mirror- Two");
        System.out.println("Touch Screen Music Player- YES");
        System.out.println("Roof Window- "+Ans);
    }
}

```

```

class Luxury extends Car_Factory
{
    String Ans;
    public void getprice(double price)
    {
        if(price>10&&price<14)
            Ans="No";
        else
            Ans="Yes";
    }
    public void detail(String company_name,String car_name)
    {
        System.out.println("Company- "+company_name);
        System.out.println("Name of Car- "+car_name);
        System.out.println("Color- Black/White/Orange/Red");
        System.out.println("Fuel- Diesel");
        System.out.println("Gears- Auto");
    }
}

```

```

}
public void accessories()
{
    System.out.println("Types of Tyres- Alloy Wheels");
    System.out.println("Airbags- YES");
    System.out.println("Back Wiper- YES");
    System.out.println("Side Mirror- Two");
    System.out.println("Touch Screen Music Player- YES");
    System.out.println("Roof Window- YES");
    System.out.println("Automotive Garbage Cans- "+Ans);
    System.out.println("Automotice Air Freshner- "+Ans);
    System.out.println("Button Start- "+Ans);
}
}

```

```

public class Main
{
    public static void main(String[] args)
    {
        Scanner scan = new Scanner(System.in);
        int ch;
        Car_Factory obj;
        while(true)
        {
            System.out.println("Which Car you want to See?- ");
            System.out.println("\n1.Small Car\n2.Sedan Car\n3.Luxary Car\n4.Exit");
            ch=scan.nextInt();
            System.out.println();
            switch(ch)
            {
                case 1:
                    obj= new Small_car();
                    obj.input();
                    obj.display(obj);
                    break;
                case 2:
                    obj= new Sedan();
                    obj.input();
                    obj.display(obj);
                    break;
                case 3:
                    obj= new Luxary();
                    obj.input();
                    obj.display(obj);
                    break;
                case 4:
                    System.out.println("\n-----");
                    return;
                default:
                    System.out.println("INVALID CHOICE !!");
                    System.out.println("\n-----");
                    break;
            }
        }
    }
}

```

}

/*OUTPUT:

Which Car you want to See?-

- 1.Small Car
- 2.Sedan Car
- 3.Luxary Car
- 4.Exit

1

Company- HYUNDAI
Car- XCENT
Rough Budget(in Lakhs)- 8

Company- HYUNDAI
Name of Car- XCENT
Color- Black/White/Orange/Red
Fuel- Petrol
Gears- Manual

Types of Tyres- Alloy Wheels
Airbags- Yes
Back Wiper- Yes
Side Mirror- Two
Touch Screen Music Player- Yes

Which Car you want to See?-

- 1.Small Car
- 2.Sedan Car
- 3.Luxary Car
- 4.Exit

3

Company- TATA
Car- NEXON
Rough Budget(in Lakhs)- 12

Company- TATA
Name of Car- NEXON
Color- Black/White/Orange/Red
Fuel- Diesel
Gears- Auto

Types of Tyres- Alloy Wheels
Airbags- YES
Back Wiper- YES
Side Mirror- Two

Touch Screen Music Player- YES

Roof Window- YES

Automotive Garbage Cans- No

Automotive Air Freshner- No

Button Start- No

Which Car you want to See?-

1.Small Car

2.Sedan Car

3.Luxary Car

4.Exit*/